MIL-C-43761C
25 March 1988
SUPERSEDING
MIL-C-43761B
28 September 1984

MILITARY SPECIFICATION

CUP, WATER CANTEEN, W/WIRE HANDLE, CORROSION-RESISTING STEEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of corrosion-resisting steel canteen cup with a folding wire handle (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-P-35	- Passivation Treatments for Corrosion-Resisting Steel
QQ-S-763	- Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting
QQ-S-766	- Steel Plates, Sheets, and Strip - Corrosion-Resisting
PPP-B-26	- Bag, Plastic, (General Purpose)
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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MILITARY

MIL-W-8611 - Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys; Process For
MIL-W-12332 - Welding, Resistance, Spot, Seam, and Projection; for Fabricating Assemblies of Low-Carbon Steel
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling for Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-9-284 - Cup, Canteen; Assembly and Details

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components. The materials and components shall be as specified herein and on Drawing 2-9-284. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document (see 6.4).

3.2.1 Steel, corrosion-resisting.

3.2.1.1 Sheet and strip. Corrosion-resisting steel sheet and strip for cup, (body and hinge), shall be class 302 or 304 of QQ-S-766.

3.2.1.2 Wire. Corrosion-resisting steel wire shall conform to condition B, class 302 or 304, form round, finish cold of QQ-S-763.

3.3 Design and construction. The design and construction of the canteen cup shall conform to Drawing 2-9-284 and the requirements specified herein (see 6.5).

3.3.1 Body. The body of the cup shall be fabricated of the corrosion-resisting steel specified in 3.2.1.1 and shall be of seamless construction conforming to the dimensions specified on Drawing 2-9-284. The top edge shall be free of sharp edges, cracks, burrs, slivers, or rough surfaces. After forming, the vertical walls of the body shall have a minimum hardness of 46 on the 30N Rockwell scale when tested as specified in 4.5.5.

3.3.2 Handle. The handle shall be fabricated of corrosion-resisting steel specified in 3.2.1.2 and shall be welded in accordance with MIL-W-8611 or MIL-W-12332. The resistance "T" welds of the handle shall withstand a minimum total force of 235 pounds when tested as specified in 4.5.4. The handle shall conform to the design, dimension, and detail requirements of Drawing 2-9-284. The handle shall fit snugly against the cup within 1/8 inch when folded and the right handle shall rest and lock into the left handle when in use.

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3.3.3 Hinge. The hinge shall be fabricated of corrosion-resisting steel specified in 3.2.1.1 and shall conform to the design, dimension, and details specified on Drawing 2-9-284. The hinge shall be attached by spot welds or projection welds in accordance with MIL-W-12332, and shall be located on the cup as shown on Drawing 2-9-284. The welds shall be free of splatter, burn holes, cracks or fractures, and discoloration and shall be tightly joined when tested as specified in 4.5.1, 4.5.2, or 4.5.3, as applicable. The hinge shall be a close fit to the contour of the body and there shall be no gaps between edge of hinge and body greater than 1/64 inch.

* 3.4 Finish. The canteen cup shall be given a uniform frosted lusterless finish approximating a commercial 2D sheet finish and passivated in accordance with QQ-P-35. There shall be no evidence of rust, staining, pitting, or spotting when tested as specified in 4.5.6.

3.5 Marking. The letters US, the year of manufacture, and the manufacturer's name or trademark shall be stamped on the hinge as shown on Drawing 2-9-284.

3.6 Workmanship. The finished canteen cup shall be clean and free of any dent or being bent out of shape, stain, ripple, fracture, die or draw mark. The handles and hinge shall also be free of any sharp edge, burr, or sliver. Indentation on the inside of cup due to welding of the hinge is acceptable providing the indentation is smooth and free from sharp edges, weld splatter, burr, crack, or burn through area, and does not exceed 0.005 inch in depth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested as specified in 4.5.2 through 4.5.6, as applicable. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. In-process examination shall be made to establish conformance to the welding on wire handles requirements of Drawing 2-9-284. Whenever nonconformance is noted, correction shall be made to the applicable process and all items processed.

4.4.2.2 In-process spot or projection weld testing. The spot or projection weld sample peel test shall be conducted on two test specimens per welding machine rather than sample cups. The spot or projection welding testing shall be performed as specified in 4.5.1 at the start of each day's shifts production. Failure of test shall require correction of the welding process.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of cups. The sample unit shall be one cup. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not a uniform, frosted, lusterless finish approximating a commercial 2D sheet finish		X
	Evidence of rust, staining, pitting, or spotting (slight water spotting, easily removed by washing shall not be considered a defect)	X	
Construction and workmanship (applicable to all components and assemblies)	Seam, lap, lamination or crease, or metal ripple	X	
	Any component missing	X	
	Component dented		X
	Component fractured	X	
	Component bent out of shape		X
	Rim of cup contains sharp edge, burr, crack, sliver, or rough spot	X	
	Evidence of die or draw mark		X
	Handle does not fold around cup within 1/8 inch as specified, or right handle does not nest into left handle	X	
	Hinge plate is not close fit against body, i.e. contains gap along edge greater than 1/64 inch (use feeler gage)		X
	Handle binds or requires force in opening or folding		X
	Handle or hinge contains sharp edge, burr, or sliver	X	
Welds	Contains crack or fracture	X	
	Not number specified	X	
	Not in location shown on drawing		X
	Contains burn hole, splatter, or discoloration	X	
	Evidence of poor fusion	X	
	Any missing weld	X	
	Weld on handle not neatly performed, i.e. contains rough or sharp projection or bead		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Welds (cont'd)	Indentation in weld area on inside of cup (Indentation on inside of the cup due to welding of the hinge is acceptable providing the indentation is smooth and free from sharp edges, weld splatter, burr, crack, burn through area, and does not exceed 0.005 inch in depth)		X
Marking	Missing, illegible, or incomplete		X

* 4.4.4 End item dimensional examination. The end items shall be examined for the dimensions annotated with an asterisk on Drawing 2-9-284. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of cups. The sample unit shall be one cup. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items shall be tested as specified in 4.5.2 or 4.5.3, 4.5.4, 4.5.5, and 4.5.6. The lot size shall be expressed in units of cups. The sample unit shall be one cup. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

* 4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

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Examine	Defect
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required Cups not arranged in container as specified

* 4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Spot or projection weld sample peel test. Each spot or projection weld test specimen shall be of the type and thickness of material used in the cup body and hinge. Spot or projection weld each section as shown on Drawing 2-9-284 and then place in a vise or jig. Bend the parent metal piece and peel the hinge piece with pliers or other suitable tool. Inability of any spot weld to pull a 5/32 inch (minimum) hole in either piece of material shall constitute failure of this test. Failures of the projection welds or lack of fusion under the projection weld shall constitute failure of the test. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion.

4.5.2 Spot weld cup peel test. Place the cup in a jig or vise and peel the hinge from the cup. Inability of each weld to leave a 5/32 inch (minimum) hole in either the cup or hinge shall constitute a defect.

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4.5.3 Projection weld cup peel test. Separate the weldment of the cup and hinge by driving a chisel between welded components in an unwelded area or by peeling the hinge back against the weld until failure occurs. Failure outside the periphery of the weld area shall be considered evidence that the welds are satisfactory. Lack of fusion under the projection weld shall constitute a defect. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion.

4.5.4 Test for resistance "T" welds of the handle. Place the cup in a jig or vise and exert a force against both "T" welds of the handle. Inability of any weld to withstand the minimum force specified in 3.3.2 shall constitute a defect.

4.5.5 Rockwell hardness test. Test the Rockwell hardness of the vertical walls of the body of the cup in accordance with ASTM E 18. Take four Rockwell hardness readings on the vertical walls each 90 degrees apart around the cup and approximately half way between the top and the bottom of the cup. Any sample cup having one or more readings less than 46 on the Rockwell 30N scale as specified in 3.3.1 shall be classified as a defect.

4.5.6 Finish test. Expose the complete assembled cup to 95-100 percent relative humidity at 72° ± 2°F for not less than 16 hours to determine compliance with rust, staining, pitting, and spotting requirements of 3.4. Inability of the cup to conform to requirements of 3.4 shall constitute a defect.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each cup, with the handles in the closed position, shall be placed in an end opening, minimum 8 inch by 12 inch polyethylene bag conforming to type II, grade A of PPP-B-26. The minimum thickness of the film shall be 0.002 inch. The end of the bag shall be secured by tucking into the cup so as to completely cover the top edges and inside bottom of the cup. Six cups shall then be nestled together in the most compact manner.

5.1.2 Commercial. The cups shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

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* 5.2.1 Level A packing. Ninety-six cups, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, grade V2s of PPP-B-636. The inside dimensions of each shipping container shall be approximately 22-1/2 inches in length, 14 inches in width, and 16 inches in depth. Approximate dimensions are furnished as a guide only. Level A unit packs shall be compactly arranged within the shipping container, 4 in length and 4 in width with the cups in an upright position. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B packing. Ninety-six cups, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636. The inside dimensions of each shipping container shall be approximately 22-1/2 inches in length, 14 inches in width, and 16 inches in depth. Approximate dimensions are furnished as a guide only. Level A unit packs shall be arranged as specified in 5.2.1. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.5.

5.2.3 Commercial. Cups, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), cups packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 32 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

6.1 Intended use. The corrosion-resisting steel cups are for use with canteen, water, plastic, MIL-C-43103. The cup is used for drinking hot and cold liquids, soups, and stews in the field.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.5 Hydroforming. Hydroforming has been found to produce acceptable cups.

6.6 Subject term (key word) listing.

Canteen
Corrosion-resisting steel
Mess kit

* 6.7 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43761C	Cup, Water Canteen, W/Wire Handle, Corrosion-Resisting Steel

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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